

Discrete Mathematics: Assignment 1

Mid Semester Answer Sheet

Assignment submitted to

Indian Statistical Institute

by

Anik Barury CS 2604



DEPARTMENT OF COMPUTER SCIENCE
INDIAN STATISTICAL INSTITUTE
203 BARRACKPORE TRUNK ROAD, KOLKATA - 700108
WEST BENGAL, INDIA

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Question 1

Problem: Let T be a tree and let S be a set of vertices of T such that every edge of T has at least one endpoint in S . Prove that T has an independent set I such that $|I| \geq |V(T)| - |S|$.

Proof: We want to prove that T has a set I such that the size of I is at least the number of vertices in T minus the size of S . Construct the set I by taking all vertices of T that are not in S :

$$I = V(T) \setminus S$$

By the definition of set complement,

$$|I| = |V(T) \setminus S| = |V(T)| - |S|$$

Since equality implies inequality, we can write,

$$|I| \geq |V(T)| - |S|$$

Showing I is an independent set:

Using proof by contradiction, let's assume I is not an independent set.

So this means $\exists$ an edge (u, v) that is adjacent such that $u, v \in I$.

Again as $u, v \in I \implies u \notin S$ and $v \notin S$ because $I = V(T) \setminus S$. $\therefore$ The edge (u, v) has neither u nor v in S . But this contradicts the problem statement, which is that every edge of T has at least one endpoint in S .

This implies our assumption was false, and I must be an independent set.

So we have shown I is an independent set and $|I| \geq |V(T)| - |S|$.

Question 2

Problem: Let $G = (V, E)$ be a graph. Suppose that every vertex of G has degree at most D . Prove that G has a proper coloring using at most $D + 1$ colors.

Proof by Induction:

Base Case: Let $n = 1$. A graph with a single vertex has degree 0. So $D = 0$. The graph can be colored with $D + 1 = 0 + 1 = 1$ color. True.

Inductive Hypothesis: Let's assume that any graph with $n - 1$ vertices and maximum degree at most D can be properly colored using at most $D + 1$ colors.

Inductive Step: Let G be a graph with n vertices and maximum degree $\Delta(G) \leq D$. Reconstruct G to make G' by removing an arbitrary vertex v (and all connected edges). So $G' = G \setminus \{v\}$. G' has $n - 1$ vertices.

As we removed edges, G' cannot have maximum degree strictly greater than $\Delta(G)$:

$$\Delta(G') \leq \Delta(G) \leq D \implies \Delta(G') \leq D$$

From the inductive hypothesis, G' with $n - 1$ vertices has a proper coloring using at most $D + 1$ colors.

Adding the removed vertex v back to G' restores G . In G , vertex v has degree at most D . $\therefore$ it has at most D neighbors.

In a proper coloring sequence, at most D colors are consumed by the D neighbors. Since we have $D + 1$ colors in our palette, there must be at least one color not used by any neighbor of v .

Assign this unused color to vertex v . Hence, we have properly colored graph G using at most $D + 1$ colors. By mathematical induction, the statement holds for all finite graphs.

Question 3

Problem: Prove that every tournament contains a vertex from which every other vertex can be reached by a directed path of length at most 2.

Proof: A tournament is a directed graph obtained by assigning exactly one direction to every pair of distinct vertices. Thus, for any two distinct vertices u, v , exactly one of $u \rightarrow v$ or $v \rightarrow u$ holds.

Since V is finite and non-empty, choose a vertex $v \in V$ of maximum out-degree, such that:

$$d^+(v) \geq d^+(x) \quad \forall x \in V$$

Let $u \neq v$ be arbitrary. Exactly one of $v \rightarrow u$ and $u \rightarrow v$ holds.

CASE 1: $v \rightarrow u$

Then u is reachable from v by a directed path of length 1.

CASE 2: $u \rightarrow v$

Here $u \notin N^+(v)$. We claim there exists some $w \in N^+(v)$ such that $w \rightarrow u$, which provides the path $v \rightarrow w \rightarrow u$ of length 2.

Suppose not; then for every $w \in N^+(v)$, the arc $w \rightarrow u$ is absent. ($N^+(v)$ is the set of out-neighbors of v). Since w is distinct from u ($u \notin N^+(v)$), the tournament property forces:

$$u \rightarrow w \quad \forall w \in N^+(v)$$

Together with the case condition $u \rightarrow v$, we obtain:

$$N^+(u) \supseteq N^+(v) \cup \{v\}$$

The union is disjoint because $v \notin N^+(v)$ (tournaments contain no self-loops).

Hence,

$$d^+(u) \geq d^+(v) + 1 > d^+(v)$$

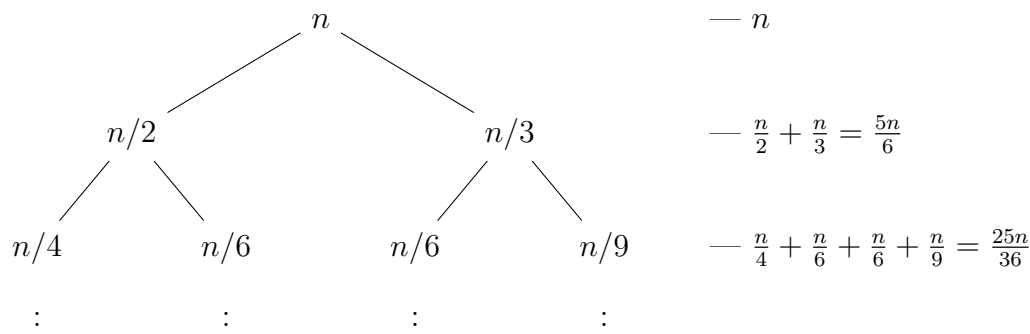
contradicting the maximality of the out-degree of v .

Therefore, such a vertex w must exist, and u is reachable from v in at most two steps. Since $u \neq v$ was arbitrary, every other vertex in T is reachable from v by a directed path of length at most 2.

Question 4(a)

Problem: Find asymptotically tight bound for $T(n) = T(\lfloor n/2 \rfloor) + T(\lfloor n/3 \rfloor) + n$.

Solving via Recursion Tree method:



$$\begin{aligned}
 \therefore T(n) &= 1 \cdot T(1) + n + \frac{5n}{6} + \frac{25n}{36} + \dots \\
 &= 1 \cdot 1 + n \left(1 + \frac{5}{6} + \frac{25}{36} + \dots \right) \\
 &= 1 + n \left[\frac{1}{1 - 5/6} \right] = 6n + 1
 \end{aligned}$$

So, $T(n) = \Theta(n)$.

Showing it recursively inductively, that $\exists c, d \mid d \cdot n \leq T(n) \leq c \cdot n$:

Showing $T(n) \leq c \cdot n$:

I.H.: Assume $T(k) \leq c \cdot k \quad \forall k < n$.

I.S.: Substituting,

$$\begin{aligned}
 T(n) &\leq c \left(\frac{n}{2} \right) + c \left(\frac{n}{3} \right) + n \\
 T(n) &\leq c \left(\frac{5n}{6} \right) + n \\
 T(n) &\leq n \left[\frac{5c}{6} + 1 \right]
 \end{aligned}$$

We want to prove $T(n) \leq cn$.

$$\begin{aligned}
 n \left[\frac{5c}{6} + 1 \right] &\leq cn \\
 \frac{5c}{6} + 1 &\leq c \\
 1 &\leq c - \frac{5c}{6} \\
 1 &\leq \frac{c}{6} \implies c \geq 6
 \end{aligned}$$

Since $\exists$ a valid constant $c \geq 6$, we conclude that $T(n) = O(n)$.

Showing $d \cdot n \leq T(n)$:

I.H.: Assume $T(k) \geq d \cdot k \quad \forall k < n$.

I.S.: Substituting,

$$\begin{aligned} T(n) &\geq d \left(\frac{n}{2} \right) + d \left(\frac{n}{3} \right) + n \\ T(n) &\geq n \left(\frac{5d}{6} + 1 \right) \end{aligned}$$

We want to prove $T(n) \geq d \cdot n$.

$$\begin{aligned} dn &\leq n \left(\frac{5d}{6} + 1 \right) \\ d &\leq \frac{5d}{6} + 1 \end{aligned}$$

Putting $d = 1$, we get $1 \leq 11/6$, which is true.

Since $\exists$ a valid constant $d = 1$, we conclude $T(n) = \Omega(n)$.

As $T(n) = O(n)$ and $T(n) = \Omega(n) \implies \mathbf{T(n) = \Theta(n)}$.

Question 4(b)

Problem: $T(n) = T(\lfloor \sqrt{n} \rfloor) + \Theta(\log n)$.

Solving via recursion tree method:

$$\begin{array}{rcl} n & \text{---} & \log_2 n \\ n^{1/2} & \text{---} & \frac{1}{2} \log_2 n \\ n^{1/4} & \text{---} & \frac{1}{4} \log_2 n \\ \vdots & \text{---} & \vdots \\ T(1) & & \end{array}$$

To find the height of the tree, we set the term to 1:

$$n^{\frac{1}{2^k}} = 1 \implies \log_2 n = 2^k \implies \log_2(\log_2 n) = k$$

$\therefore$ Height of the tree is $\log_2(\log_2 n)$.

Summing the work at each level:

$$\begin{aligned} T(n) &= T(1) \cdot 1 + \log_2 n + \frac{1}{2} \log_2 n + \frac{1}{4} \log_2 n + \dots \\ &= 1 \cdot 1 + \log_2 n \left(1 + \frac{1}{2} + \frac{1}{4} + \dots \right) \\ &= 1 + \log_2 n \left[\frac{1}{1 - 1/2} \right] = 1 + 2 \log_2 n = \Theta(\log n) \end{aligned}$$

Showing it recursively (inductively):

$\exists c, d \mid d \log_2 n \leq T(n) \leq c \log_2 n$.

Showing $T(n) \leq c \log_2 n$:

I.H.: Assume $T(k) \leq c \log_2 k \quad \forall k < n$.

I.S.: Substituting and using $k_2 \log_2 n$ for the upper limit of the extra work:

$$\begin{aligned} T(n) &\leq c \log_2(\sqrt{n}) + k_2 \log_2 n \\ T(n) &\leq c \left(\frac{1}{2} \log_2 n \right) + k_2 \log_2 n \\ T(n) &\leq \log_2 n \left(\frac{c}{2} + k_2 \right) \end{aligned}$$

We want to prove $T(n) \leq c \log_2 n$:

$$\frac{c}{2} + k_2 \leq c \implies k_2 \leq \frac{c}{2} \implies c \geq 2k_2$$

$\therefore k_2$ is a constant, we can pick a c that is twice as large as k_2 . Thus $T(n) = O(\log n)$.

Showing $T(n) \geq d \log_2 n$:

I.H.: Assume $T(k) \geq d \log_2 k \quad \forall k < n$.

I.S.: We use $k_1 \log_2 n$ for the lower limit of the extra work:

$$T(n) \geq d \log_2(\sqrt{n}) + k_1 \log_2 n$$

$$T(n) \geq d \left(\frac{1}{2} \log_2 n \right) + k_1 \log_2 n$$

$$T(n) \geq \log_2 n \left(\frac{d}{2} + k_1 \right)$$

We want to show $T(n) \geq d \log_2 n$:

$$\frac{d}{2} + k_1 \geq d \implies k_1 \geq \frac{d}{2} \implies d \leq 2k_1$$

Similarly, $T(n) = \Omega(\log n)$ holds good.

So, as $T(n) = O(\log n)$ and $T(n) = \Omega(\log n)$, we have $\mathbf{T(n)} = \mathbf{\Theta(\log n)}$.

Question 5(a)

Problem: A binary string is called legal if between any two consecutive 1's there are at least two 0's. Let a_n be the number of legal binary strings of length n . Derive a recurrence for a_n , together with appropriate initial conditions.

Solution:

Consider the termination structure of an arbitrary valid string of length n :

CASE 1: String ends in 0

Removing the trailing 0 leaves any legal binary string of length $n - 1$.

This yields a_{n-1} possibilities.

CASE 2: String ends in 1

A trailing 1 must be preceded by at least two 0's, locking the suffix into $\dots 001$. Removing these three fixed bits leaves any legal string of length $n - 3$.

This yields a_{n-3} possibilities.

Because every legal string terminates in either 0 or 1, we obtain:

$$a_n = a_{n-1} + a_{n-3} \quad \text{for } n \geq 3$$

Base Cases:

$$a_0 = 1 \quad (\text{empty string})$$

$$a_1 = 2 \quad (0, 1)$$

$$a_2 = 3 \quad (00, 01, 10)$$

Consistency check: $a_3 = a_2 + a_0 = 3 + 1 = 4$ (valid strings: 000, 001, 010, 100).

Question 5(b)

Problem: Find $A(x)$ as a rational function where $A(x) = \sum_{n \geq 0} a_n x^n$.

Solution:

Multiplying $a_n = a_{n-1} + a_{n-3}$ by x^n and summing over $n \geq 3$:

$$\begin{aligned} \sum_{n=3}^{\infty} a_n x^n &= \sum_{n=3}^{\infty} a_{n-1} x^n + \sum_{n=3}^{\infty} a_{n-3} x^n \\ \sum_{n=0}^{\infty} a_n x^n - a_0 - a_1 x - a_2 x^2 &= x \sum_{n=3}^{\infty} a_{n-1} x^{n-1} + x^3 \sum_{n=3}^{\infty} a_{n-3} x^{n-3} \\ A(x) - 1 - 2x - 3x^2 &= x[A(x) - a_0 - a_1 x] + x^3 A(x) \\ A(x) - 1 - 2x - 3x^2 &= x[A(x) - 1 - 2x] + x^3 A(x) \\ A(x) - xA(x) - x^3 A(x) &= 1 + 2x + 3x^2 - x - 2x^2 \\ A(x)[1 - x - x^3] &= 1 + x + x^2 \\ A(x) &= \frac{1 + x + x^2}{1 - x - x^3} \end{aligned}$$

Question 5(c)

Problem: Use your answer to determine the exponential growth rate of a_n .

Solution:

The exponential growth rate of the sequence is governed by the dominant singularity of $A(x)$, which is the root of the denominator with the smallest absolute value:

$$1 - x - x^3 = 0 \implies x^3 + x - 1 = 0$$

Let α be the unique real root of $x^3 + x - 1 = 0$. The growth rate is given by $1/\alpha$.

Solving $f(x) = x^3 + x - 1 = 0$ via the Newton-Raphson method ($x_{n+1} = x_n - \frac{x_n^3 + x_n - 1}{3x_n^2 + 1}$):

$$x_0 = 1.0$$

$$x_1 = 0.75$$

$$x_2 \approx 0.6860$$

$$x_3 \approx 0.6823$$

Hence $\alpha \approx 0.6823$, and the exponential growth rate is approximately $1/0.6823 \approx 1.4656$.

Question 6(a)

Problem: For $n \geq 0$, let a_n be the number of integer solutions of

$$x_1 + x_2 + x_3 + x_4 = n$$

satisfying $x_1, x_2, x_3, x_4 \geq 0$, where:

- x_1 is even
- x_2 is NOT divisible by 3
- $x_3 \leq 5$
- $x_4 \geq 2$

Find the ordinary generating function $A(x) = \sum_{n \geq 0} a_n x^n$.

Solution:

Translating the individual restrictions into generating functions:

- x_1 is even:

$$1 + x^2 + x^4 + \dots = \frac{1}{1 - x^2}$$

- x_2 is not divisible by 3:

$$\begin{aligned} (1 + x + x^2 + \dots) - (1 + x^3 + x^6 + \dots) &= \frac{1}{1 - x} - \frac{1}{1 - x^3} \\ &= \frac{(1 + x + x^2) - 1}{1 - x^3} = \frac{x(1 + x)}{1 - x^3} \end{aligned}$$

- $x_3 \leq 5$:

$$1 + x + x^2 + x^3 + x^4 + x^5 = \frac{1 - x^6}{1 - x}$$

- $x_4 \geq 2$:

$$x^2 + x^3 + x^4 + \dots = \frac{x^2}{1 - x}$$

Multiplying these components gives:

$$A(x) = \left(\frac{1}{1 - x^2} \right) \left(\frac{x(1 + x)}{1 - x^3} \right) \left(\frac{1 - x^6}{1 - x} \right) \left(\frac{x^2}{1 - x} \right)$$

Question 6(b)

Problem: Simplify $A(x)$ to a rational function.

Solution:

Factoring the terms:

$$A(x) = \frac{1}{(1-x)(1+x)} \cdot \frac{x(1+x)}{1-x^3} \cdot \frac{(1-x^3)(1+x^3)}{1-x} \cdot \frac{x^2}{1-x}$$

Canceling $(1+x)$ and $(1-x^3)$:

$$A(x) = \frac{x^3(1+x^3)}{(1-x)^3} = \frac{x^3 + x^6}{(1-x)^3}$$

Question 6(c)

Problem: Determine a_{10} .

Solution:

Rewrite $A(x)$ using the negative binomial series:

$$A(x) = (x^3 + x^6)(1-x)^{-3}$$

By the Extended Binomial Theorem:

$$(1-x)^{-3} = \sum_{k=0}^{\infty} \binom{k+3-1}{3-1} x^k = \sum_{k=0}^{\infty} \binom{k+2}{2} x^k$$

We extract the coefficient of x^{10} by finding:

- Coefficient of x^7 in $(1-x)^{-3}$ for the x^3 term ($k=7$):

$$\binom{7+2}{2} = \binom{9}{2} = \frac{9 \times 8}{2} = 36$$

- Coefficient of x^4 in $(1-x)^{-3}$ for the x^6 term ($k=4$):

$$\binom{4+2}{2} = \binom{6}{2} = \frac{6 \times 5}{2} = 15$$

Summing both components gives:

$$a_{10} = 36 + 15 = \mathbf{51}$$